

LAKHANLAL

Postdoctoral Researcher | Computational Materials Science

Email: lankhanlall@campus.technion.ac.il | lakhanlal.nitk@gmail.com

Phone: +972-58-361-9247 (Israel) | +91-7320002375 (India)

[ORCID: 0000-0002-3582-3854](#) | [Google Scholar](#) | [Website](#)

Department of Materials Science and Engineering, Technion - Israel Institute of Technology, Haifa 3200003, Israel

RESEARCH AREAS

Computational materials science; density functional theory and electronic-structure calculations; functional metal oxides and metal-organic frameworks; electrocatalysis and photocatalysis; oxygen evolution reaction and water splitting; electronic conductivity in electrocatalysts; non-equilibrium Green's function methods; solid oxide electrolysis and electrochemical energy conversion.

SKILLS

Computational methods: VASP; SIESTA; Python for scientific computing; density functional theory; electronic-structure analysis.

Electrochemistry and cell testing: Cyclic voltammetry, linear sweep voltammetry, electrochemical impedance spectroscopy, and solid oxide electrolysis cell testing for steam/CO₂ co-electrolysis.

Materials characterization: X-ray diffraction, FTIR spectroscopy, X-ray photoelectron spectroscopy, scanning electron microscopy, Raman spectroscopy, surface-area analysis, and gas chromatography.

Instrumentation: Fuel-cell/electrolysis-cell test stations, potentiostat/galvanostat, impedance analyzer, source meter, dilatometer, gas chromatograph, high-temperature gas-tight furnaces, and pellet press.

Materials synthesis: Solution combustion, co-precipitation, citrate complexation, and synthesis of metal-oxide electrode and electrolyte materials.

EDUCATION AND QUALIFICATIONS

Year	Qualification	Institution	Result
2007	Matriculation	Bihar School Examination Board, Patna, India	70.40%
2010	Certificate in Automobiles & Farm Machinery	Sant Longowal Institute of Engineering & Technology (SLIET), Longowal, Punjab, India	74.75%
2012	Diploma in Industrial & Production Engineering	Sant Longowal Institute of Engineering & Technology (SLIET), Longowal, Punjab, India	73.35%
2015	Bachelor of Engineering (B.E.) in Mechanical Engineering (Welding Technology)	Sant Longowal Institute of Engineering & Technology (SLIET), Longowal, Punjab, India	7.39/10 CGPA
2019	Master of Technology (M.Tech.) in Energy Technology	Tezpur University, Assam, India	8.64/10 CGPA; Distinction
2022-2026	Ph.D. in Materials Science and Engineering <i>Thesis: Computational Study of Water Oxidation Reaction Activity and Electronic Conductivity in Transition Metal-Doped Nickel Oxyhydroxide Electrocatalysts</i>	Technion - Israel Institute of Technology, Haifa, Israel	GPA: 94.7

POSTDOCTORAL RESEARCH EXPERIENCE

Postdoctoral Researcher

01 July 2026-Present

Department of Materials Science and Engineering, Technion - Israel Institute of Technology, Haifa, Israel

Studying Zeolites for CO₂ to Fuel conversion

PROFESSIONAL RESEARCH EXPERIENCE

Senior Research Fellow

August 2019-February 2022

Department of Chemical Engineering, National Institute of Technology Karnataka, India

Research on development and demonstration of solid oxide electrolysis cell technology for co-electrolysis of carbon dioxide and water for syngas production, including cell testing and development of electrode and electrolyte materials.

TEACHING EXPERIENCE (AS TA)

Course	Term	Level
Advanced Project in Materials Engineering (3150014)	Autumn 2024	Undergraduate
Advanced Project in Materials Engineering (315015)	Spring 2024	Undergraduate
Introduction to Computational Materials Science (315057)	Spring 2024	Undergraduate

FELLOWSHIPS, AWARDS AND HONORS

Year	Fellowship / Award
2025	KLA Excellence Award.
2024	Kenneth and Gloria Levy Graduate Fellowship for Excellence.
2023	Short-Term Scientific Mission (STSM) Grant, COST Action 18234, for a research visit to TU Wien for research on metal-organic frameworks.
2019	Tezpur University Research and Innovation Grant for the project "Surface-modified oxygen reduction reaction catalyst derived from sulfur-rich coal and rice straw for fuel-cell application" (six months).
2017-2019	MHRD Postgraduate Scholarship for M.Tech. studies at Tezpur University.

GATE EXAM DETAILS

1. GATE 2017, Mechanical Engineering (ME), Score 430/1000
2. GATE 2019, Engineering Sciences (XE) (Thermodynamics and Materials Science), Score 533/1000

PUBLICATIONS

Peer-Reviewed Journal Articles

3. Michal Mizrahi; Rawnaq Batheesh; **Lakhanlal**; Wenjamin Moschkowitsch; David A. Cullen; Yoed Tsur; Maytal Caspary Toroker ; Lior Elbaz, "Development of Ternary Transition Metal Oxide Catalysts for Oxygen Evolution Reaction", *ACS Applied Energy Materials* (2026), [DOI](#)
4. Bischoff, J.; von Baeckmann, C.; Naghdi, S.; Ertl, A.; Vorobyev, V.; Naryshkina, A.; **Lakhanlal**; Kählig, H.; Kronlachner, L.; Woodward, R. T.; Kleitz, F.; Limbeck, A.; Caspary Toroker, M.; Morris, A. J.; Eder, D.. "Photocatalytic Ammonia Synthesis Using Fe-Based MOFs: The Role of Ligand Functionalization." *Journal of the American Chemical Society* (2026). [DOI](#)
5. Abbas, L.; **Lakhanlal**; Bhowmick, S.; Batheesh, R.; Elbaz, L.; Caspary Toroker, M.; Tsur, Y.. "Toward Efficient Hydrogen Production: Impact of Solid Solution of Tungsten on Nickel-Iron Hydroxide OER Catalysts." *ACS Catalysis* 16(5), 4449-4462 (2026). [DOI](#)
6. Ottapilakkil, F.; Shirasangi, R.; **Lakhanlal**; Dasari, H. P.; Saidutta, M. B.. "Electrochemical Performance of CO₂ Electrolysis in an Electrolyte-Supported (NiO-YSZ/NiO-SDC/ScSZ/LSCF-GDC/LSCF) Solid Oxide Electrolysis Cell." *Discover Electrochemistry* 2, Article 9 (2025). [DOI](#)
7. **Lakhanlal***; Rimon, O.*; Moschkowitsch, W.; Cohen Taguri, G.; Elbaz, L.; Caspary Toroker, M.. "Effect of Heat Treatment on Improving OER Activity of NiFeOOH-Based Aerogels: A Combined Experimental and Theoretical Study." *Molecular Catalysis* 561, 114164 (2024). [DOI](#)
8. Shirasangi, R.; **Lakhanlal**; Dasari, H. P.; Saidutta, M. B.. "Current-Voltage (i-V) Characteristics of an Electrolyte-Supported (NiO-YSZ/NiO-SDC/ScSZ/LSCF-GDC/LSCF) Solid Oxide Electrolysis Cell during CO₂/H₂O Co-Electrolysis." *Chemical Physics Impact* 9, 100670 (2024). [DOI](#)
9. Ayala, P.; Naghdi, S.; Nandan, S. P.; Myakala, S. N.; Rath, J.; Saito, H.; Guggenberger, P.; **Lakhanlal Lakhanlal**; Kleitz, F.; Caspary Toroker, M.; Cherevan, A.; Eder, D.. "The Emergence of 2D Building Units in Metal-Organic Frameworks for Photocatalytic Hydrogen Evolution: A Case Study with COK-47." *Advanced Energy Materials* 13, 2300961 (2023). [DOI](#)
10. Srijith; **Lakhanlal**; Dasari, H. P.; Saidutta, M. B.. "Electrical Conductivity Studies on LAMOX-Based Electrolyte Materials for Solid Oxide Fuel Cells." *Ceramics International* 48(19B), 29229-29237 (2022). [DOI](#)
11. Das, A.; **Lakhanlal**; Shajahan, I.; Dasari, H. P.; Saidutta, M. B.; Dasari, H.. "Dilatometer Studies on LAMOX-Based Electrolyte Materials for Solid Oxide Fuel Cells." *Materials Chemistry and Physics* 258, 123958 (2021). [DOI](#)
12. Rajvanshi, K.; Patil, S. S.; **Lakhanlal**; et al.. "Promotional Effect of Nickel Addition on Soot Oxidation Activity of Ce_{0.9}Pr_{0.1}O₂ Oxide Catalysts." *Chemical Papers* 74, 4581-4592 (2020). [DOI](#)

* Equal contribution.

Review Articles

13. **Lakhanlal**; Caspary Toroker, M.. "Filling the Gaps on the Relation between Electronic Conductivity and Catalysis of Electrocatalysts for Water Splitting Using Computational Modelling." *Current Opinion in Electrochemistry* (2023). [Publisher page](#)

14. Sarmah, S.; **Lakhanlal**; Kakati, B. K.; Deka, D.. "Recent Advancement in Rechargeable Battery Technologies." *WIREs Energy and Environment* e461 (2022). [DOI](#)

Refereed Conference Proceedings

15. **Lakhanlal**; Kakati, B. K.. "Synthesis of Nitrogen- and Sulfur-Nitrogen-Co-Doped Carbon-Based Oxygen Reduction Reaction Catalyst from Rice Straw and Reeling Waste of Muga Silk (*Antheraea assamensis*)." *Materials Today: Proceedings* (2021). [DOI](#)
16. **Lakhanlal**; Dasari, H. P.; Saidutta, M. B.. "Shrinkage Behavior, Thermal Expansion Behavior, and Electrical Conductivity Study of Samarium-Doped Ceria Electrolytes." *ECS Transactions* 103(1), 1239-1250 (2021). [DOI](#)

Submitted Manuscript

17. Singh, S.; **Lakhanlal Lakhanlal**; Albo, A.; Bernstein, J. B.. "A Density Functional Theory Study of the Formation, Interaction, and Electronic Activity of Carbon and Oxygen Defects at the 4H-SiC(0001)/SiO₂ Interface." *Materials Today Communications* (submitted).

CONFERENCE PRESENTATIONS

Oral Presentations

1. **2026: Electronic Conductivity Study of Transition-Metal-Doped NiOOH-Based OER Electrocatalysts.** The Young RSCC Student Conference, Technion.
2. **2021: Synthesis of Nitrogen- and Sulfur-Nitrogen-Co-Doped Carbon-Based Oxygen Reduction Reaction Catalyst from Rice Straw and Reeling Waste of Muga Silk (*Antheraea assamensis*).** International Conference on Materials Science and Mathematics for Advanced Technology (MSMAT 2021), NITTE, Karnataka, India.

Poster Presentation

3. **2023: Study of Heat-Treated gamma-NiFeOOH Aerogel for Oxygen Evolution Reaction Using DFT.** International Conference on Development of Nanocrystal Materials through Computational Modelling, Technion, Haifa, 18 July 2023.

CONFERENCE PRESENTATIONS

1. Prof. Maytal Caspary Toroker, (PhD and Postdoctoral research advisor), Professor at Department of Materials science and Engineering, Technion-Israel Institute of Technology, maytalc@technion.ac.il, (O) +972-73-378-4298
2. Prof. Hari Prasad Dasari, (Worked with him in a DST project as an SRF), Professor at Department of Chemical engineering, National Institute of Technology, Karnataka, Surathkal, energyhari@nitk.edu.in, (M) +919177304950
3. Dr. Biraj Kumar Kakati, (M.Tech. Thesis advisor), bkakati@tezu.ernet.in, Assistant Professor at the Department of Energy, Tezpur University, Assam. (O) +913712275313